

C Programming Language:

Basics Of C

1. History and Features of C
2. Importance of C
3. About Procedural Language
4. Role of Compiler
5. Role of Interpreter
6. The Structure of a C Program
7. Writing C Program
8. Building an Executable Version of a C Program
9. Debugging a C Program
10. Examining and Running a C Application Program

An Overview Of C And Buzzwords

1. Data Type, Variable
2. Operators
3. Control Statement
4. Preprocessor Directives Input and Output Operators
5. Reading/Writing Characters
6. Formatted input/output Function
- 7.

Control Statement

1. The IF.....ELSE Statement
2. IF.....ELSE ladder
3. Nesting of IF.....ELSE Statements
4. The Switching Statements
5. The do-while Statement
6. The while statement
7. FOR Statements

Array In C

1. Array : What and Why?
2. One Dimensional Arrays
3. Two Dimensional Arrays
4. Multi Dimensional Arrays
5. Dynamic Arrays



String Handling

1. Reading String from Terminal
2. Writing String to Screen
3. String Handling Functions
4. Table of String

Function In C

1. Function Basics
2. Advantage of Function
3. Recursion
4. Variable Storage Classes
5. Variable arguments Function

Pointers In C

1. Understanding Pointers
2. Pointer expressions
3. Pointer and Arrays
4. Pointers and Character String
5. Pointers to Functions
6. Pointers and Structures

Structure and Unions

1. Defining a Structure
2. Advantage of Structure
3. Size of Structure
4. Arrays of Structures
5. Structures and Functions
6. Defining Unions

Dynamic Memory Allocation

1. Introduction to Dynamic Memory Allocation
2. Malloc
3. Calloc
4. Realloc
5. Free

Linked-List



1. Concepts of Linked Lists
2. Creating a Linked Lists
3. Inserting/Deleting an item
4. Application of Linked Lists

File Management In C

1. Introduction to File Management
2. Opening/Closing a File
3. Input/Output operations on Files
4. Error Handling During I/O Operations
5. Command Line Arguments

Preprocessor

1. Macro Substitution
2. File Inclusion
3. Computer Control Directives

Project-Development

C++ Syllabus

C++ Overview

1. C++ Characteristics
2. Object-Oriented Terminology
3. Polymorphism
4. Object-Oriented Paradigm
5. Abstract Data Types
6. I/O Services
7. Standard Template Library
8. Standards Compliance

C++ Functions and Variable

1. Functions: Declaration and Definition
2. Variables: Definition, Declaration, and Scope
3. Variables: Dynamic Creation and Derived Data
4. Arrays and Strings in C++
5. Qualifiers

Classes in C++



1. Defining Classes in C++
2. Classes and Encapsulation
3. Member Functions
4. Instantiating and Using Classes
5. Using Constructors
6. Multiple Constructors and Initialization Lists
7. Using Destructors to Destroy Instances
8. Friendship

Operator Overloading

1. Operator Overloading
2. Working with Overloaded Operator Methods

Initialization and Assignment

1. Initialization vs. Assignment
2. The Copy Constructor
3. Assigning Values
4. Specialized Constructors and Methods
5. Constant and Static Class Members

Storage Management

1. Memory Allocation
2. Dynamic Allocation: new and delete

Inheritance

1. Overview of Inheritance
2. Defining Base and Derived Classes
3. Constructor and Destructor Calls

Polymorphism

1. Overview of Polymorphism

Input and Output C++ Programs

1. Standard Streams
2. Manipulators
3. Unformatted Input and Output
4. File Input and Output

Exception



1. Exceptions
2. Inheritance and Exceptions
3. Exception Hierarchies

4. Inside an Exception Handler

Templates

1. Template Overview
 2. Customizing a Templated Method
 3. Standard Template Library Containers
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